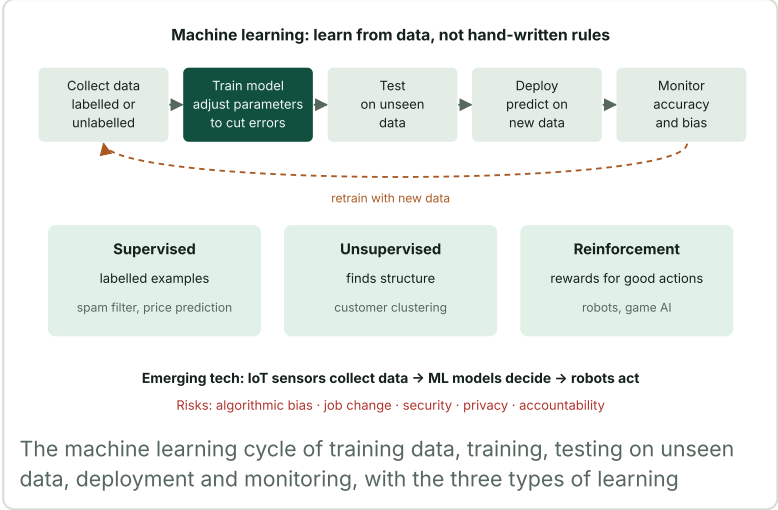
Artificial intelligence, machine learning and emerging technologies

IS16 FACT FILE

Specification: students should be able to

- Describe artificial intelligence and machine learning, including supervised, unsupervised and reinforcement learning
- Describe emerging technologies including the Internet of Things and robotics
- Evaluate the impact of these technologies on organisations

Key words: artificial intelligence, machine learning, training data, supervised learning, unsupervised learning, reinforcement learning, neural network, algorithmic bias, Internet of Things, sensor, robotics, automation



The machine learning cycle of training data, training, testing on unseen data, deployment and monitoring, with the three types of learning

AI and machine learning

Artificial intelligence is the ability of computer systems to perform tasks that normally require human intelligence. Machine learning is a branch of AI in which algorithms learn from data rather than following explicitly programmed rules. A model is trained on training data, tested on unseen data, then deployed to make predictions. Supervised learning uses labelled data, unsupervised learning finds patterns in unlabelled data, and reinforcement learning learns by trial and error using rewards.

Emerging technologies

The Internet of Things connects physical objects such as meters, vehicles and medical monitors so they can collect and exchange data automatically. Robotics uses programmable machines to sense and act in the physical world, from factory assembly to warehouse picking and surgery. Combined with AI, these technologies enable automation, predictive maintenance and real-time decision-making across many sectors, including agriculture, health, transport and retail.

Impact on organisations

AI and emerging technologies can increase productivity, reduce costs and errors, support better decisions and create new services. However, they can displace workers, require costly investment and new skills, and introduce risks: biased or unexplainable decisions, security weaknesses in IoT devices, and privacy concerns from large-scale data collection. Organisations must comply with data protection law, keep humans responsible for important decisions, and plan for staff retraining.

T Key terms from the mark schemes

Artificial intelligence (AI)
The ability of computer systems to perform tasks that normally require human intelligence, such as recognising images, understanding language and making decisions.

Machine learning
A branch of AI in which algorithms improve their performance at a task by learning patterns from data rather than following explicitly programmed rules.

Supervised learning
Machine learning in which a model is trained on labelled examples whose correct outputs are known.

Algorithmic bias
Systematic unfairness in the outputs of an AI system, usually caused by biased or unrepresentative training data.

Internet of Things (IoT)
A network of everyday physical objects with embedded sensors, processors and connectivity that collect and exchange data.

Exam focus

Extended answers on AI and emerging technologies reward depth and application. Define the technology precisely, give specific uses in the organisation named, and weigh benefits against drawbacks for different stakeholders – the organisation, employees, customers and society. Avoid science-fiction claims; current AI is narrow. Use technical terms such as training data, algorithmic bias and automation correctly, and finish with a justified conclusion rather than a summary.

? Check yourself

1. Define machine learning. [2]
2. Distinguish between supervised and unsupervised learning, giving an example of each. [4]
3. Explain what is meant by algorithmic bias and how it can occur. [2]

Answers: 1. Machine learning is a type of AI in which algorithms improve their performance at a task by learning from data (1) / rather than being explicitly programmed with rules – the model finds patterns in training data and uses them to make predictions on new data (1). 2. Supervised learning trains the model on labelled data, i.e. inputs with the correct outputs (1) / example: emails labelled spam/not spam to classify new emails; predicting house prices (1) / Unsupervised learning finds patterns or structure in unlabelled data with no correct answers given (1) / example: clustering customers into segments for marketing (1). 3. Algorithmic bias is when an AI system makes unfair or discriminatory decisions that systematically favour or disadvantage particular groups (1) / it can occur when the model is trained on biased or unrepresentative data, e.g. a recruitment model trained on past hiring that favoured men (1). Accept: biased choice of features or design by developers.